Business Process Explanation: SACCO Loan Lifecycle and Arrears Monitoring

Course: INSY 8311 – Database Development with PL/SQL
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Phase I

1. Business process definition

Process: SACCO loan lifecycle, from loan application to repayment, arrears monitoring and loan closure.

Purpose and objectives: to give or refuse loans using clear rules, to record every instalment accurately, and to detect late payments early so that officers can act before a loan becomes a default.

Expected outcome: each approved loan has a schedule and a full repayment history, overdue instalments raise alerts automatically, and managers can see the portfolio at risk.

Problem addressed: many SACCOs track loans on paper or spreadsheets, so late payments are found only when a loan is seriously overdue, and loans can be approved for members who already have unpaid balances.

Why it matters: late and unrecovered loans reduce the funds available to other members and threaten the SACCO′s savings. Early warning and enforced rules protect members′ money.

2. Actors and responsibilities

Actor	Type	Responsibility
Member	External user	Requests loans, nominates guarantors, pays instalments
Loan Officer	Employee	Records applications, requests guarantees, follows up overdue members

Actor	Type	Responsibility
Guarantor	External user	Accepts or declines to guarantee a loan
Credit Committee	Unit	Reviews the application and risk score, approves or rejects
Cashier	Employee	Disburses funds, receives and records payments
System (APEX app + PL/SQL)	Other system	Checks arrears, scores risk, creates schedules, applies payments, raises alerts, writes the audit log
Management	Unit	Reviews portfolio at risk, decides on collections, provisioning and write-offs
Auditor / Regulator	External stakeholder	Receives portfolio reports and checks the audit trail

How the process begins

The process starts when a member needs a loan and submits a request. The Loan Officer records the application in the APEX app (create), and the System retrieves the member's record to check for unresolved arrears. If arrears exist, the System records a rejection, the officer notifies the member, and the process ends. Otherwise the officer asks the guarantor to accept the guarantee.

How it progresses and what each actor does

The **Guarantor** accepts or declines; a guarantee record is created only on acceptance. If the guarantor declines, the member may nominate another one or the application is rejected. The **System** then calculates a risk score from repayment history, loan size and guarantor cover. The **Credit Committee** reviews the application and score and records its decision. If approved, the System creates the loan and its repayment schedule, and the **Cashier** disburses the funds and records the release date. The **Member** repays by instalments: the payment is received and recorded, the System applies it to the oldest unpaid instalment, handles partial, over and early payments, and writes an audit entry. A fully paid loan is closed and archived.

Arrears detection runs separately from payments. A timer triggers a check at each due date plus a grace period. If an instalment is unpaid, the System raises one alert and moves it through the 30, 60 and 90-day buckets. The Loan Officer follows up; either the member pays and the alert is resolved, or the case is escalated to **Management**, who decide on default or write-off. The timer stops once the loan is closed or written off.

How decisions are made

Most decisions are rule-based and enforced in PL/SQL: whether arrears exist, whether a payment is normal, partial, over or early, whether an instalment is overdue, and whether the loan is fully paid. Judgement decisions stay with people: the guarantor's acceptance, the Credit Committee's approval, and Management's write-off decision.

How information moves

Information moves between actors as the application form, guarantee request, rejection notice, funds and disbursement notice, payment, arrears alert and portfolio report. Each task that touches data is marked in the diagram as create (C), retrieve (R) or update (U): applications, guarantees, risk scores, loans, schedules, repayments, alerts and audit entries are created; instalments, alert status and loan status are updated; member records and arrears are retrieved.

Support for operations and management

For daily operations, the process gives officers one reliable record of each loan and blocks unsafe actions, such as lending to a member in arrears. For management, it produces portfolio-at-risk and arrears by bucket, so managers can plan collections and provisioning, and external auditors or regulators receive a portfolio report backed by the audit log.

Data-generating activities (C = create, R = retrieve, U = update)

Activity	Mark	Information generated or used
Record application	C / R	Creates the application (member, product, amount, date); retrieves member and product details

Activity	Mark	Information generated or used
Check unresolved arrears	R	Retrieves the member's loans and open alerts
Record rejection decision	U	Updates the application decision and reason
Confirm guarantee	C	Creates the guarantor record and amount guaranteed
Calculate risk score	R / C	Retrieves repayment history; creates the risk assessment (score and band)
Review application and score	R	Retrieves the application, guarantors and score
Record committee decision	U	Updates the application decision and date
Create loan and schedule	C	Creates the loan and its instalments (due dates, amounts)
Disburse funds	U	Updates the loan with the release date and status
Record repayment	C	Creates the repayment (amount, date, payer)
Apply payment to oldest instalment	U	Updates instalment status; handles surplus and early settlement
Record audit log entry	C	Creates the audit record (table, action, old and new values, user, time)
Check instalment paid (timer)	R	Retrieves instalments due, with the grace period
Raise or update arrears alert	C / U	Creates the alert, then updates the 30, 60 or 90-day bucket
Resolve alert	U	Marks the alert resolved when the overdue instalment is cleared
Record follow-up outcome	U	Updates the alert or loan with the officer's result
Mark loan defaulted or write off	U	Updates the loan status after Management approval
Close and archive loan	U	Sets the status to closed and archived; the history is kept

Activity	Mark	Information generated or used
Retrieve PAR and arrears by bucket	R	Retrieves portfolio-at-risk figures for Management and auditors

Delete: no records are physically deleted. Closed and written-off loans are archived by status so that the repayment history and audit trail remain available.